

**SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)**  
Department of Computer Science and Engineering

**ASSESSMENT TOOL 1 - ANALYTICAL PROBLEM SOLVING**

|                  |                                                                                                                                               |               |                                                |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------|------------------------------------------------|
| Course Code:     | CSA0309                                                                                                                                       | Course Title: | Data Structures                                |
| CO Assessed:     | CO2 - Manipulation (BL3, BL4)                                                                                                                 | Assessment:   | Assessment Tool 1 - Analytical Problem Solving |
| Weightage:       | 30% of CIA                                                                                                                                    | Total Marks:  | 100                                            |
| CO2 Statement:   | Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques. |               |                                                |
| Student Name:    | PAWAN.C                                                                                                                                       | Reg. No.:     | 192521202                                      |
| Section / Batch: |                                                                                                                                               | Date:         | 17/07/2026                                     |

**Code of Conduct**

I PAWAN.C / 192521202 (Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: PAWAN.C

**Instructions**

- This test contains 80 Analytical problem-solving questions. Total: 100 Marks.
- When a student answer using an Analytical problem-solving, in evaluation the answer should be with following five requirements: Understanding of problem, select appropriate data structure, Design the solution, analyze, Clarity of representation.
- Each student should write 2 questions. No negative marking. Unanswered questions carry zero marks.
- No DTP printouts or electronic devices permitted. They should provide written materials.

| Section / Topic                              | Focus                           | Q Nos | Marks            |
|----------------------------------------------|---------------------------------|-------|------------------|
| Linked Data Structure, Search Data Structure | List Operations, Binary Search  | 1     | 50               |
| Stack Data Structure, Queue Data Structures  | Stack Notations, Priority Queue | 2     | 50               |
| <b>GRAND TOTAL:</b>                          |                                 |       | <b>100 Marks</b> |

|    |           |                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----|-----------|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 26 | 192411179 | PAPPUSANI<br>HARSHAVARDHAN | <ol style="list-style-type: none"> <li>Identify if the given expressions are balanced or not.<br/>(i) <math>[9 * (k+8 - \{ 6 + e \} + 77) / 100]</math>, (ii) <math>((a*b) + [u/2] - d)</math>.</li> <li>Given a stack with repeated elements, design an approach to compress it (value &amp; frequency) while maintaining stack operations. Apply stack ADT modifications. Analyze trade-off between time and space</li> </ol>                                                                                                                                                                                                                                                       |
| 27 | 192572144 | PAUL DAVID R               | <ol style="list-style-type: none"> <li>A circular queue has a size of 6. Execute the following operations: Enqueue(10), Enqueue(20), Enqueue(30), Dequeue(), Enqueue(40), Enqueue(50), Enqueue(60), Enqueue(70). Draw the queue after each operation and identify the values of the front and rear pointers.</li> <li>Design a system that dynamically chooses between linear search and binary search based on: <ul style="list-style-type: none"> <li>➤ Data size</li> <li>➤ Sortedness</li> <li>➤ Frequency of queries</li> </ul> Analyze decision criteria mathematically </li> </ol>                                                                                             |
| 28 | 192521202 | PAWAN C                    | <ol style="list-style-type: none"> <li>Insert the following elements 100, 200, 300, 400, 500 into the linked stack and pop the elements 100 and 200 from the linked stack. Show the linked structure before and after linked stack operations.</li> <li>An online food delivery service receives the following orders: 01, 02, 03, 04. While 02 is being prepared, an express order 05 arrives.<br/><i>Questions:</i> <ul style="list-style-type: none"> <li>➤ Explain why a normal queue may not satisfy customer requirements.</li> <li>➤ Recommend an appropriate queue implementation and justify your choice.</li> </ul> </li> </ol>                                             |
| 29 | 192524056 | PRENA M                    | <ol style="list-style-type: none"> <li>Construct an efficient system using two stacks within a single array structure to maximize memory utilization. Compare fixed partitioning and dynamic allocation strategies, and analyze how each approach affects overflow conditions, space efficiency, and operational complexity under varying workloads.</li> <li>A queue-based ticketing system serves customers in the order of arrival, but VIP customers must be prioritized occasionally. Analyze how modifying a simple queue into a priority-based queue affects fairness, waiting time, and computational complexity. Compare both systems using analytical reasoning.</li> </ol> |
| 30 | 192511341 | PUGAZHARASAN K             | <ol style="list-style-type: none"> <li>Convert the following infix expression into postfix notation. Show the stack contents after processing each symbol.<br/><math>((A+B)*C-(D-E)*(F+G))/H</math></li> <li>For the expression. <math>((8+4) \times (7-3) / (2+1)) + (5 \times (6-2))</math></li> </ol>                                                                                                                                                                                                                                                                                                                                                                              |



## Linked Stack.

A stack follows LIFO (Last in, First out) Principle.

Elements ingested : 100, 200, 300, 400, 500

Before POP:

TOP  $\rightarrow$  ~~500~~  $\rightarrow$  400  $\rightarrow$  300  $\rightarrow$  200  $\rightarrow$  100  
 $\rightarrow$  NULL

To remove 100 and 200 the elements above them must be removed first

POP operations

POP 500

POP 400

POP 300

POP 200

POP 100

After POP:

TOP  $\rightarrow$  NULL

Thus linked stack always removes the most recently inserted element first

## 2. online Food delivery - Queue

A normal queue follows FIFO (First in, First out)

orders received:

FRONT  $\rightarrow$  01  $\rightarrow$  02  $\rightarrow$  03  $\rightarrow$  04  $\rightarrow$  REAR

When express order 05 arrives a normal queue places it at the rear

01  $\rightarrow$  02  $\rightarrow$  03  $\rightarrow$  04  $\rightarrow$  05

Therefore order 05 has to wait behind the normal order which is not suitable for an express order

FRONT  $\rightarrow$  05 (Express)  $\rightarrow$  03  $\rightarrow$  04  $\rightarrow$  REAR

*Urging* The priority queue gives higher priority to express order 05 allowing it to be processed before normal orders.